

Incompatible density claims in conjecture 00000008487

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Both versions assert that the locked-function set is dense G_δ and that its complement is dense and open. These assertions cannot hold simultaneously in any nonempty topological space. The contradiction does not depend on the locking convention, measure or fractal dimension.

Topological obstruction. Let Y be a nonempty topological space and $L \subseteq Y$ be dense. If $Y \setminus L$ is open and nonempty, density of L implies

$$L \cap (Y \setminus L) \neq \emptyset,$$

which is impossible. Hence an open complement of a dense set must be empty. But an empty set cannot be dense in a nonempty space: its closure is empty. We have therefore proved

$$\neg(L \text{ dense} \wedge Y \setminus L \text{ open} \wedge Y \setminus L \text{ dense}).$$

This argument is valid without a Baire-space assumption, and even without the additional G_δ assertion.

Application to the actual interval function space. Take the genuine interval $I = [0, 1] \subset \mathbb{R}$ and its continuous observable space $Y = C(I, \mathbb{R})$, with its usual uniform topology. This space is nonempty, since the constant zero function belongs to it. Fix any interval map, for example the continuous identity map $T(x) = x$, and let L be its actual set of locked observables according to the intended definition. Since the obstruction holds for *every* subset of Y , it holds for this actual L . Thus L cannot satisfy the two asserted density clauses simultaneously. This disproves the conjecture as written, without imposing a definition of locking that might alter the statement.

A dense G_δ set can certainly have a dense complement; the added *openness* of that complement is what causes the contradiction. No conclusion about the separate Lebesgue-measure or box-dimension claims is needed. The proof remains valid for other nonempty spaces of observables and any topology on them.

Lean correspondence. The accompanying project proves the obstruction for every actual nonempty topological space and every subset, using Mathlib's `Dense` and `IsOpen` predicates. It specializes to `ContinuousMap (Set.Icc (0:Real) 1) Real`, supplies the actual zero observable, and proves that no subset can satisfy the complete claimed topology conjunction, including the G_δ predicate. This universal statement covers the actual locked-function set without an assumed certificate or a replacement dynamics model. The project is pinned to Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`.